

Density of split matrix

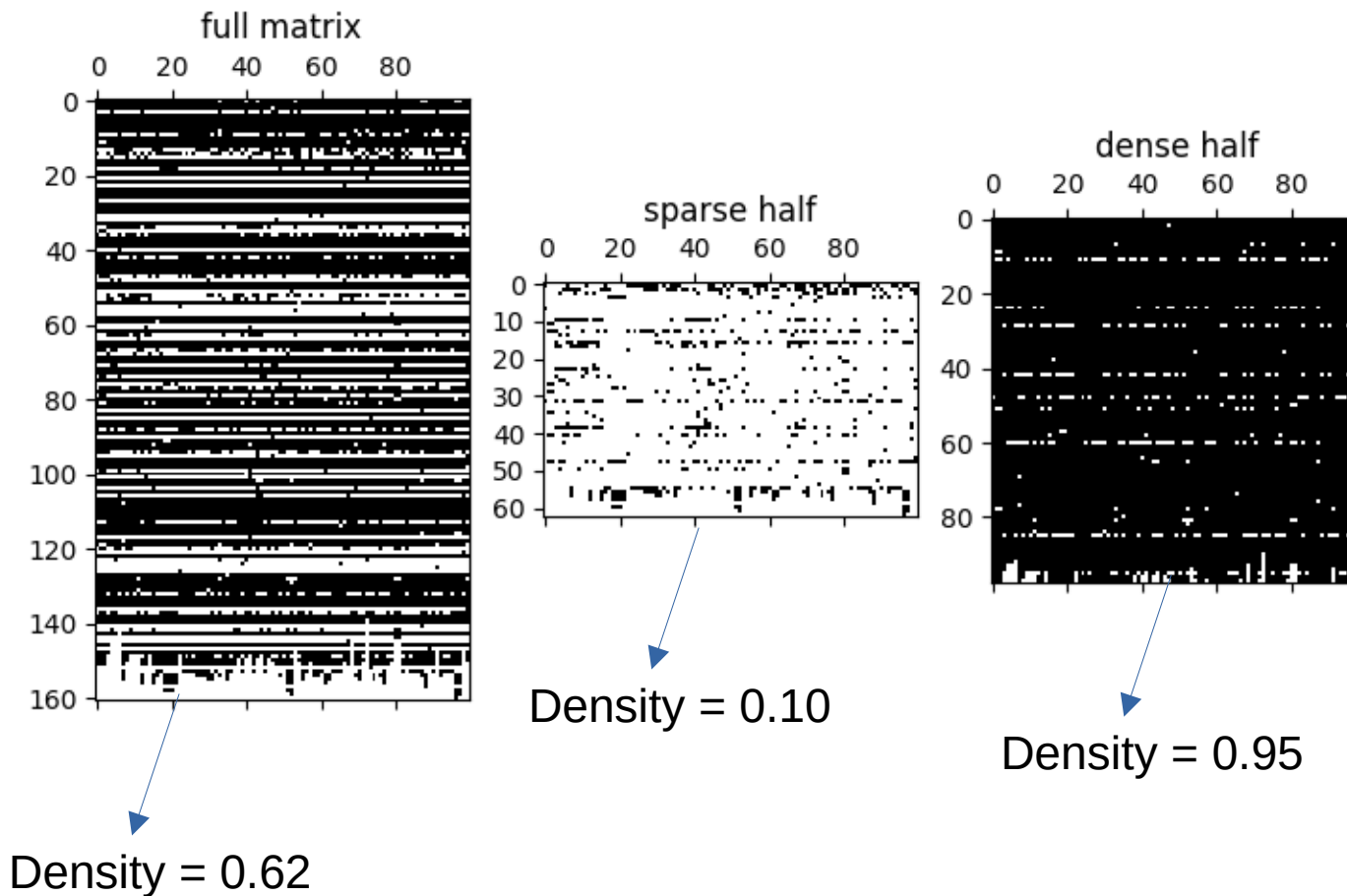
Marie Picard

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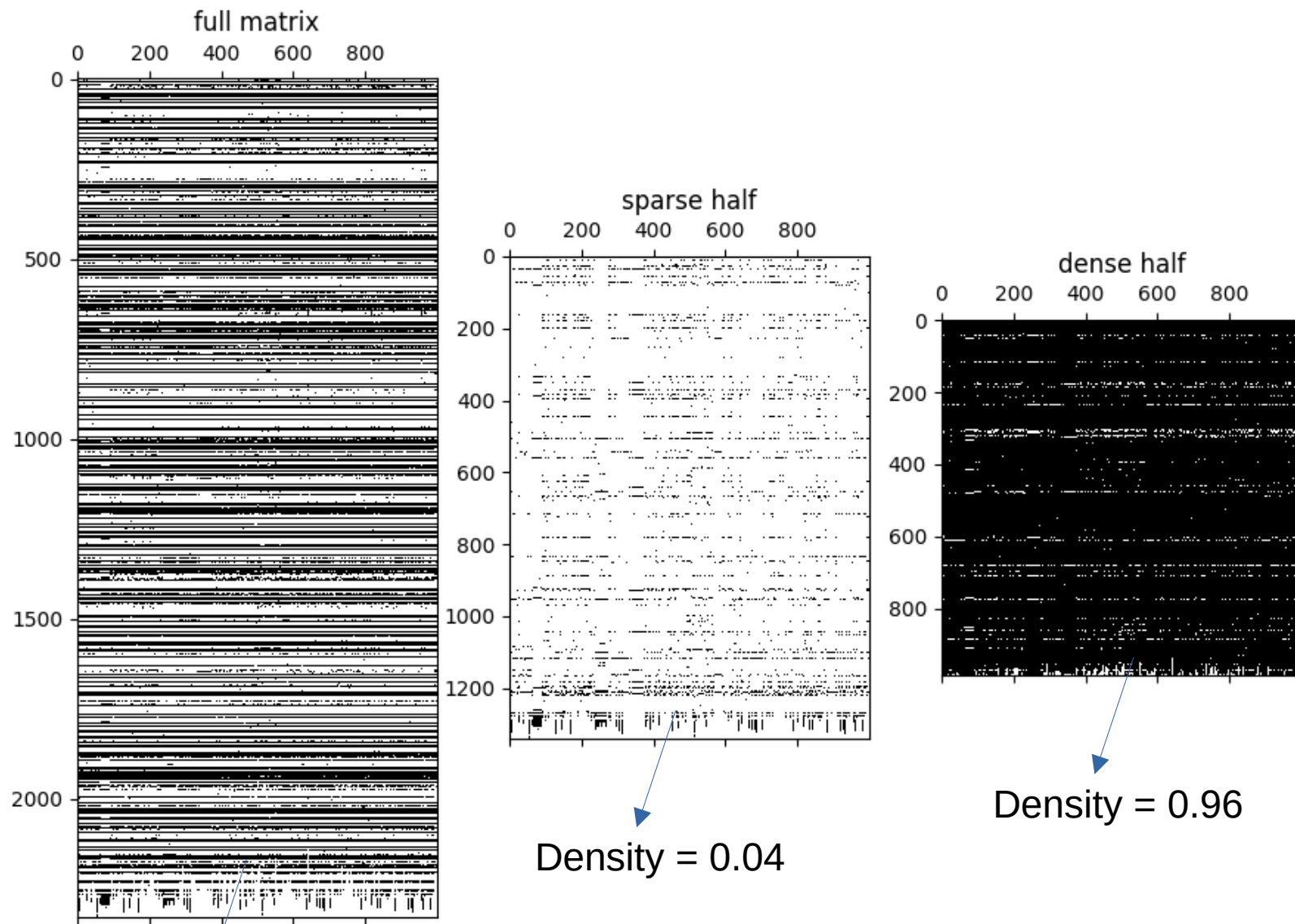
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February 6, 2026

Quick recap of past results

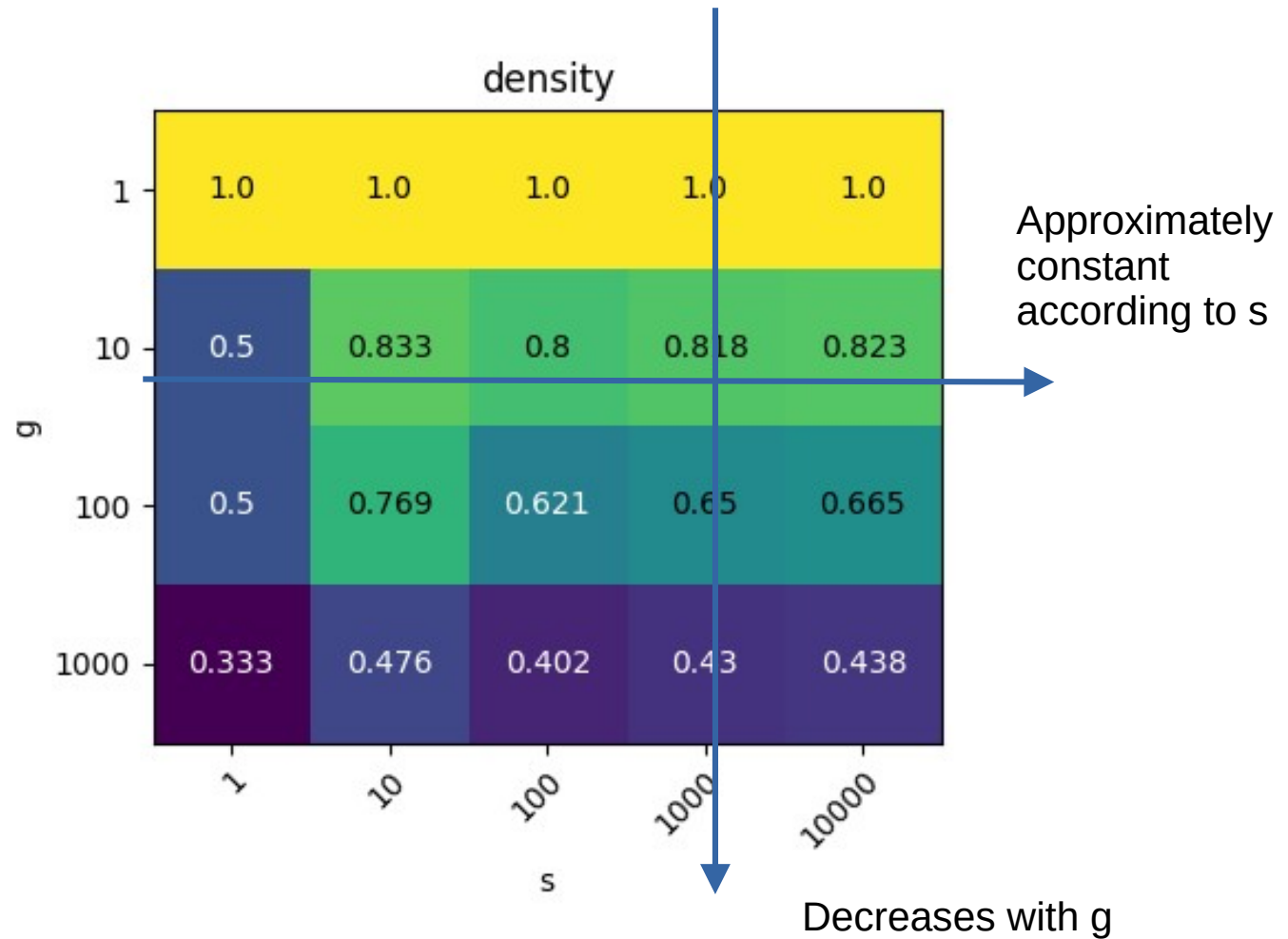


Quick recap of past results

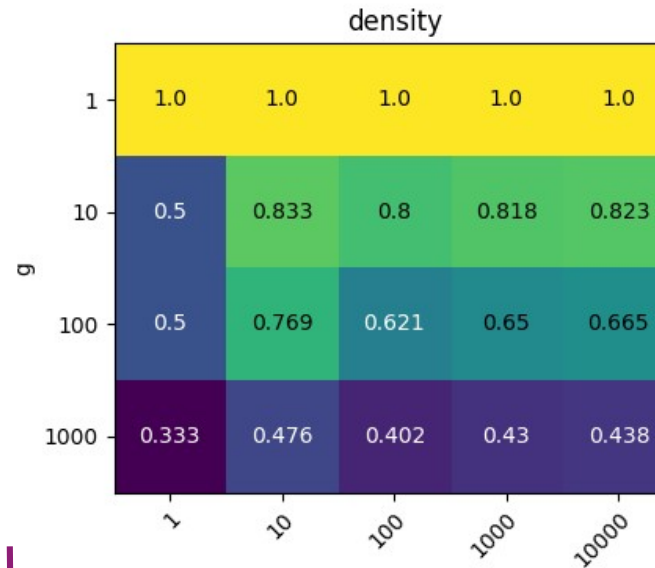


Ngono
s = 1000
g = 1000 leftmost genomes

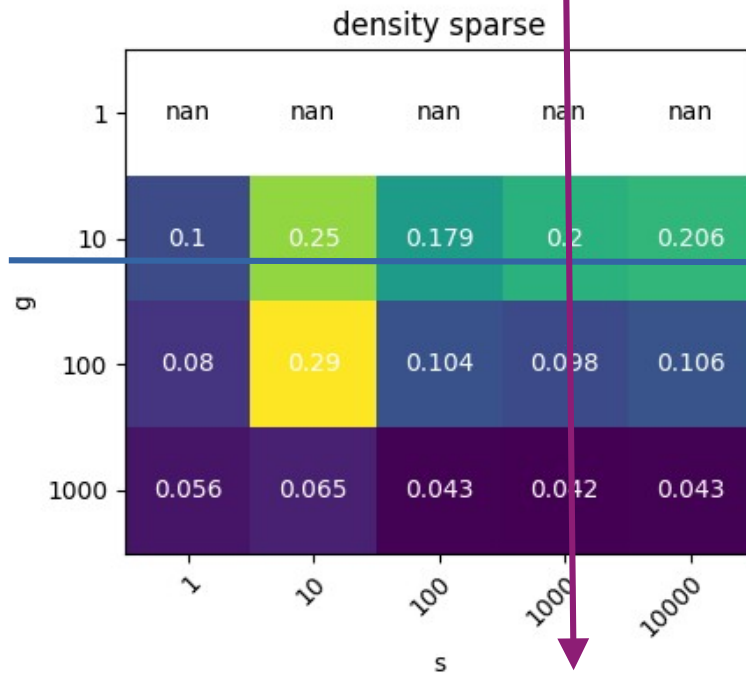
Densities full matrix



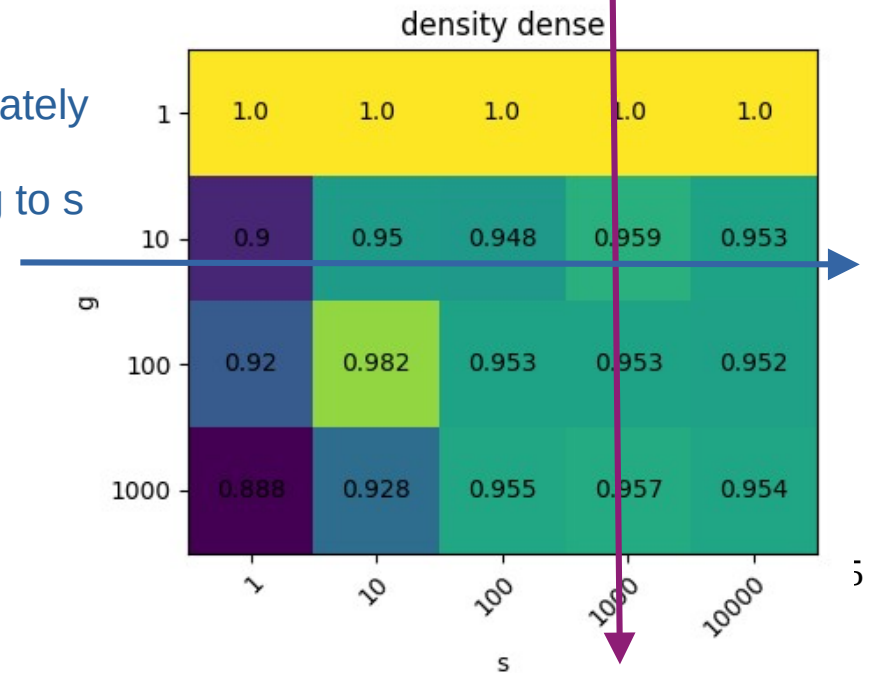
Densities split matrix



Decreases with g

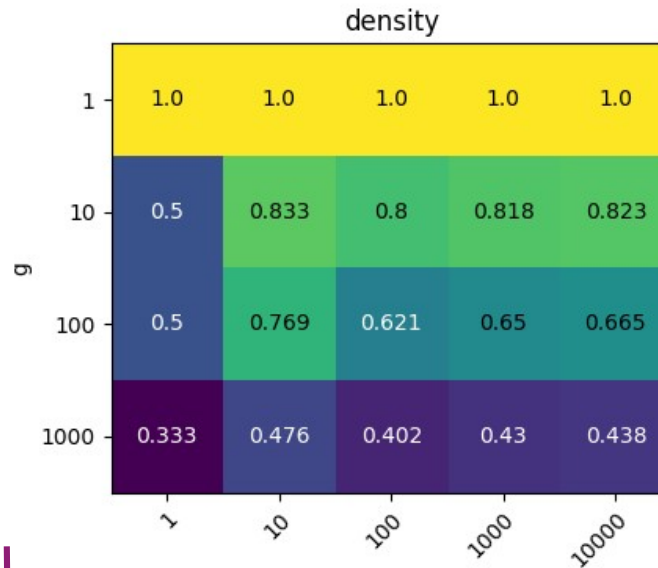


Approximately
constant
according to s

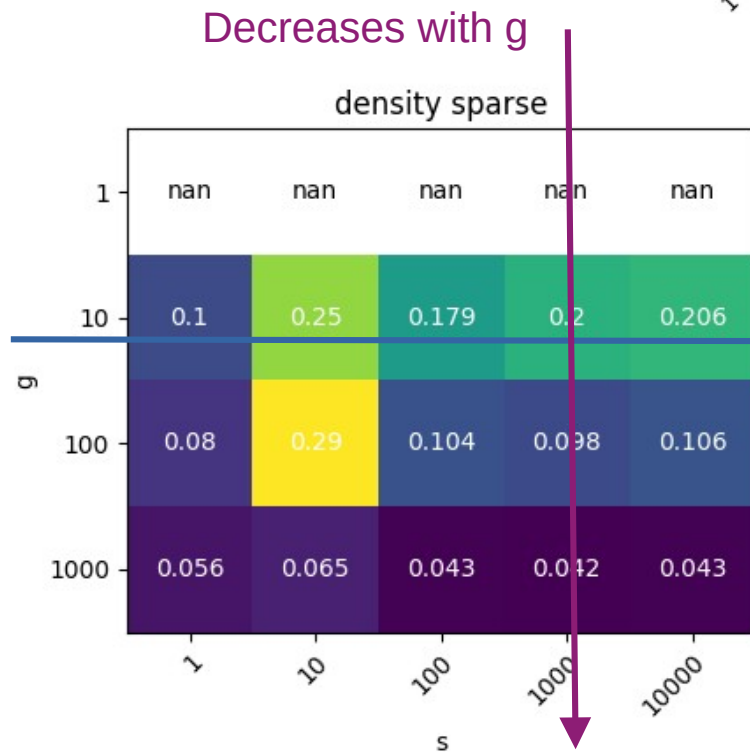


Constant with g

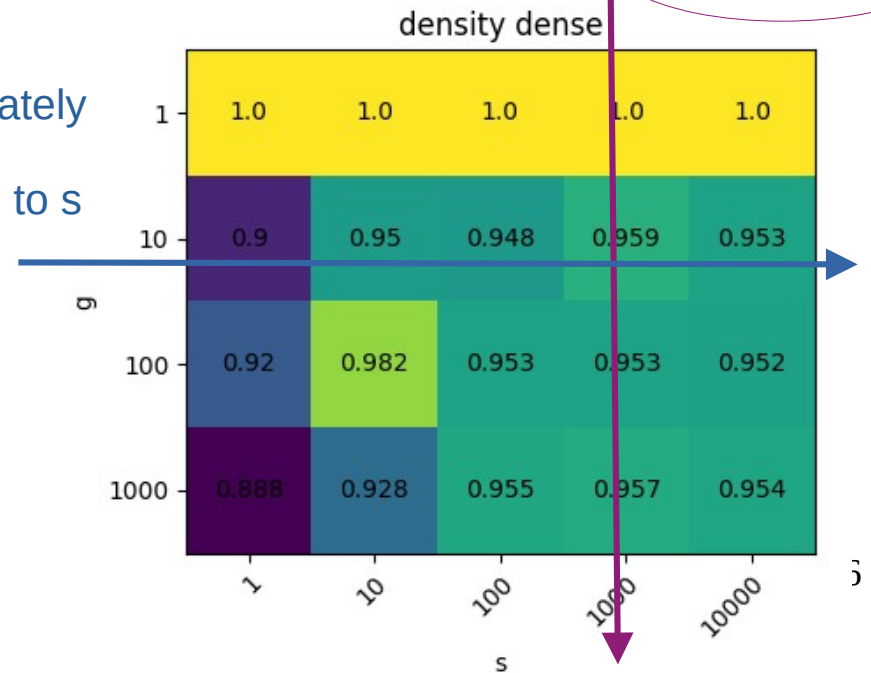
Densities split matrix



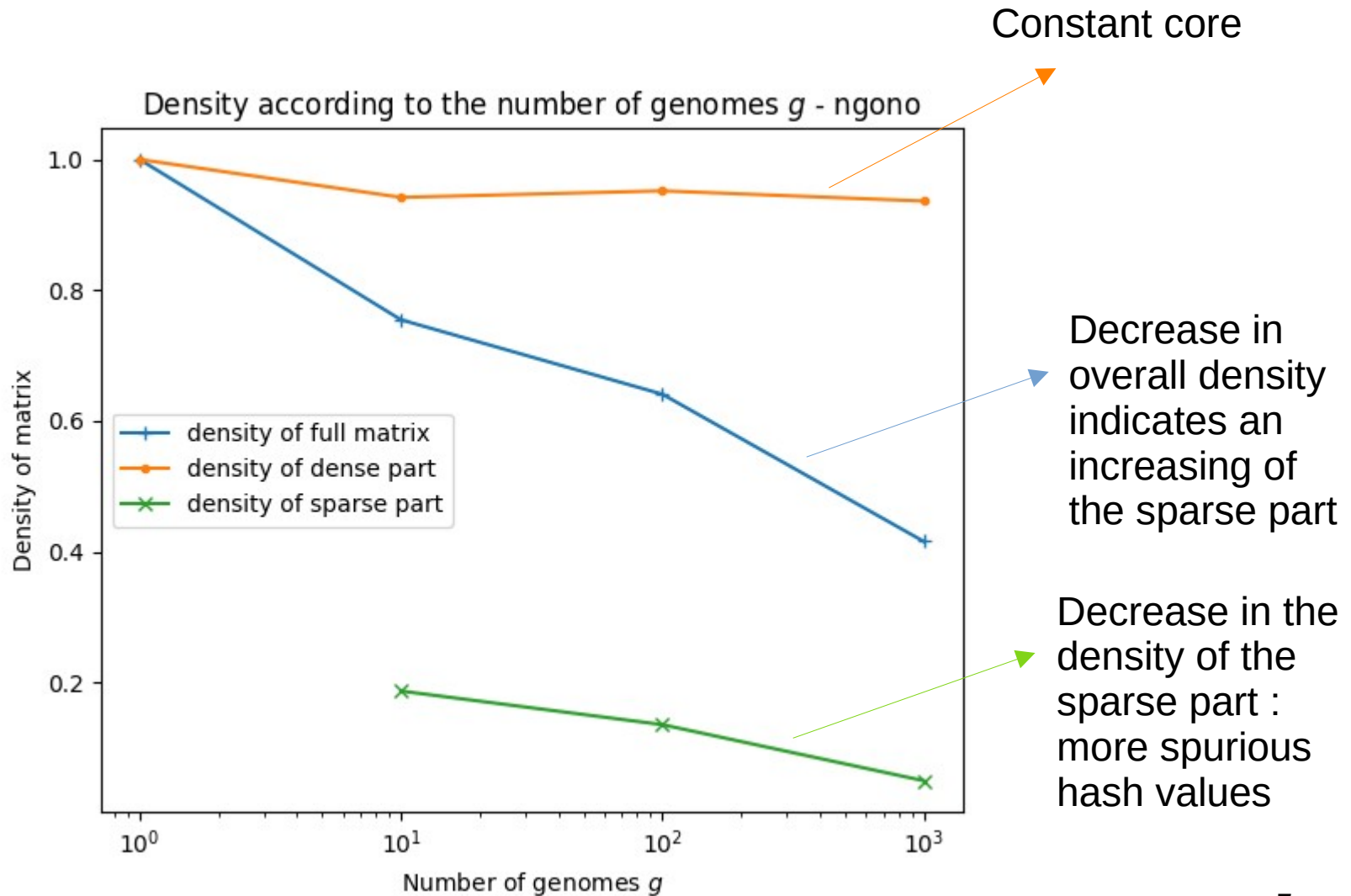
Conjecture : a “core” of hashes that corresponds to the dense part (whose number of rows doesn’t change much with g), and a part of spurious hashes



Approximately constant according to s



Densities split matrix



Conclusion

Conclusions :

- Density of sparse and dense parts of the matrix are close to extremes
- The dense part probably does not change much when g increases, and increases linearly with s : indicating a “core” of hash values

To do :

- Benchmark density and compressibility for split matrices for different species
- Keep on working on implementation of a tool
- Look into the compressibility of such matrices